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VIP Lithium-Ion Batteries Very Important Paper

# Lithium Hydride in the Solid Electrolyte Interphase of Lithium-Ion Batteries as a Pulverization Accelerator of Silicon

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**Abstract:** As a highly promising next-generation high-specific capacity anode, the industrial-scale utilization of micron silicon has been hindered by the issue of pulverization during cycling. Although numerous studies have demonstrated the effectiveness of regulating the inorganic components of the solid electrolyte interphase (SEI) in improving pulverization, the evolution of most key inorganic components in the SEI and their correlation with silicon failure mechanisms remain ambiguous. This study provides a clear and direct correlation between the lithium hydride (LiH) in the SEI and the degree of micron silicon pulverization in the battery system. The reverse lithiation behavior of LiH on micron silicon during de-lithiation exacerbates the localized stress in silicon particles and contributes to particle pulverization. This work successfully proposes a novel approach to decouple the SEI from electrochemical performance, which can be significant to decipher the evolution of critical SEI components at varied battery anode interfaces and analyze their corresponding failure mechanisms.

## Introduction

The rapid progress of the electric vehicle industry has heightened the necessity for lithium-ion batteries (LIBs) characterized by enhanced energy densities.[1] Within this domain, micron silicon has emerged as a promising candidate for substituting conventional graphite anode materials, owing to its remarkable theoretical capacity (achieving 3579 mAh $g^{-1}$ for $Li_{15}Si_4$ at ambient temperatures) as well as the abundance of natural resources available.[2] However, the utilization of silicon's ultrahigh-specific capacity often accompanies severe pulverization because of the substantial volumetric variations (up to 300 % during operation).[3] As a result, there is a substantial depletion of active lithium and active materials, which ultimately severely compromises the cycling stability of silicon.[4]

The investigation into micron silicon pulverization holds significant practical importance in enhancing cycling stability and promoting the extensive utilization of silicon. Extensive research has firmly established a strong correlation between the pulverization of micron silicon and the variation of the SEI on the anode.[2c,3,5] In the realm of lithium-ion battery systems, it is well known that the SEI with intricate structure and diverse composition assumes a critical role in impeding electronic transport and facilitating lithium-ion conduction in the interface of the anode. This has a profound impact on numerous aspects of battery performance, encompassing energy density, cycle life, calendar life, rate capability, low/high-temperature, and safety performance.[6] It has been reported that the unstable SEI which progressively grows along the nano-porous pathways on the silicon gradually encapsulates the silicon domain, disrupting the electronic conduction pathways within the electrode and resulting in silicon deactivation and capacity loss in the battery.[5a] After extensive exploration, optimization of SEI components through the design of electrolytes or artificial SEI has garnered considerable recognition for the effectiveness in controlling micron silicon pulverization.[2c,5b,7] More specifically, and it is widely accepted by the researchers that increasing the content of high modulus inorganic components (such as LiF) in SEI to mechanically stabilize the SEI on Si is an effective strategy to alleviate pulverization.[2c,8]

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However, it has been freshly discovered in our study that the degree of pulverization is actually positively correlated with the LiH content of in the SEI despite the LiH (136.15 GPa) has higher modulus than LiF (125.67 GPa),[9] which raises questions about the widely accepted SEI design principles. While the significance of SEI components in influencing micron silicon pulverization has been acknowledged by empirical evidence, there seems to be a lack of clear understanding regarding the relationship or mechanisms between the critical components within the SEI and the issue of silicon pulverization. This lack of understanding is particularly evident for lithium hydride (LiH), which has been discovered in recent years. Enhancing the understanding and exploration in this regard would contribute to a more precise and scientific delineation of the ambiguous SEI design principles. This would effectively overcome the bottleneck of scientific and rational design of the SEI, thereby accelerating the practical development progress of silicon-based anodes. In our previous research, it has been revealed that LiH is unstable and can react with the active material in germanium alloys, exacerbating the failure of the germanium alloy anode.[10] However, there is no report concerning the intricate link between LiH evolution and the degradation of silicon alloy anode. herein, we put forth a plausible hypothesis that the evolutionary behavior of LiH on silicon anodes may be intricately linked to anode failure, a phenomenon that has yet to be investigated or explored. Deepening our understanding of the LiH on silicon and the interconnected mechanisms is imperative to facilitate the design of SEI with enhanced capabilities to inhibit pulverization, thereby significantly bolstering the cycling stability of silicon-based anodes.

In this study, an unequivocal correlation has been observed between the presence of LiH in the anode and the degree of silicon pulverization in the battery system. Herein, we leverage a fusion of diverse characterization techniques, inclusive of online mass spectrometry titration and both in situ transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) to investigate the evolution of LiH and its consequential correlation with alterations in particle pulverization. The quantitative observation based on online mass spectrometry titration unfolds the dynamic evolution of LiH accumulation within the SEI during the lithiation process, as well as its subsequent depletion during the de-lithiation process. Importantly, a novel mechanism is postulated based on the comprehensive analysis using in situ TEM and HRTEM. It is suggested that the de-lithiated silicon undergoes reverse lithiation induced by the highly reactive LiH, exacerbating localized inhomogeneous lithiation and poten-tially accelerating to micron silicon pulverization or fracture. This study offers fresh insights into the mechanism behind the failure of silicon anodes, which significantly diverges from the prevailing understanding of high energy density anode

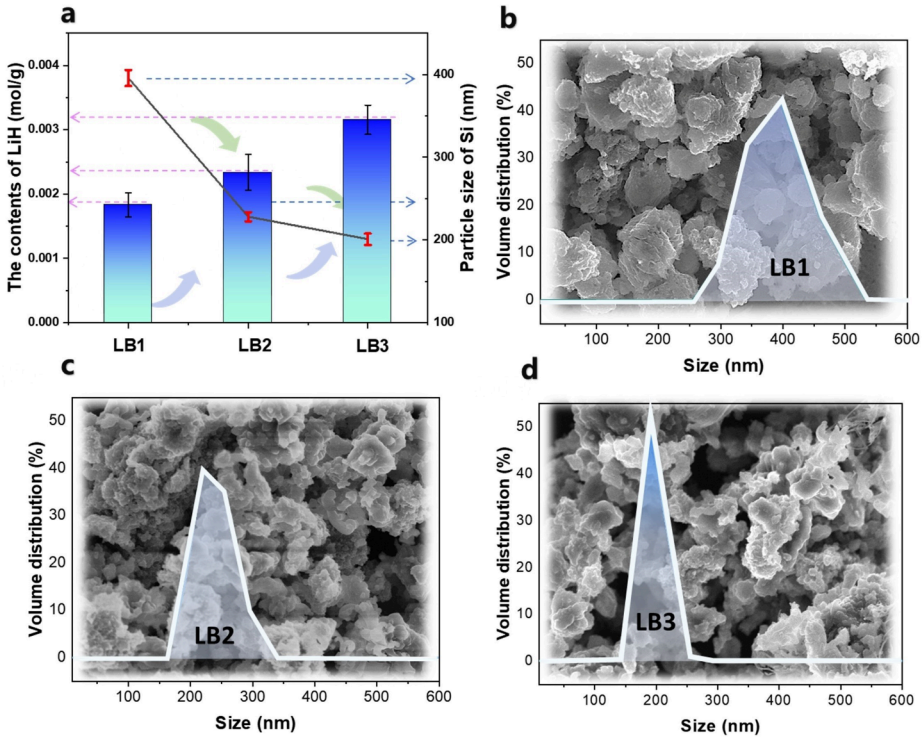



***Figure 1.*** **Morphology and the LiH amounts of cycled Si particles**. (a) The amounts of LiH (left) as well as the average particle size (right) of the silicon electrode which obtained from the Si–Li half-cell with different electrolytes (cycled 1st). (b) The typical SEM image of the Si electrode which obtained from the Si–Li half-cell (cycled 1st) with LB1 electrolyte. (c) The typical SEM image of the Si electrode which obtained from the Si–Li half-cell (cycled 1st) with LB2 electrolyte. (d) The typical SEM image of the Si electrode which obtained from the Si–Li half-cell (cycled 1st) with LB3 electrolyte. The drawing of the error bars in the figure is based on the electrode pieces in three independent battery after cycled ($n=3$).


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failure. Simultaneously, it establishes clear principles for the scientific design of stable SEI on micron silicon. Furthermore, it expands the range of approaches for investigating failure mechanisms in other anode materials.

## Results and Discussion

The assembled silicon (1–5 μm)-based pouch cells (24 mg/cm$^2$, ~2.5 Ah) exhibited poor electrochemical performance,-notably characterized by an astonishing irreversible capacity loss after the first cycle (Figure S1). Post-analysis of the disassembled cells revealed severe pulverization of the silicon anode. Furthermore, after disassembling the cells which were assembled using three different electrolytes (LB1, LB2, and LB3; respectively), we have observed a certain correlation between the content of LiH in the anode and the degree of pulverization in silicon-based batteries (Figure 1a). The obtained silicon electrodes (cycled 1 time) were subjected to mass spectrometry titration to determine the LiH content. It was evident that the electrode applied with LB2 and LB3 exhibited higher LiH content compared LB1. The trend observed in the average particle size of the three electrodes aligns impeccably with the inverse correlation found in the LiH content within the corresponding SEI. Remarkably, as the LiH content within the SEI escalates, there is a simultaneous reduction in the average particle size of the electrode following cycling, indicating an intensification of silicon particle fragmentation. Corresponding SEM results also revealed that the pulverization state of the silicon electrodes in the LB1, with lower LiH content, was significantly superior to that of the two groups with higher LiH (Figure 1b–1d).

To comprehensively investigate and elucidate the evolution behavior of LiH during the cycling of silicon anodes, a quantitative titration method employing a sensitive on-line gas analysis mass spectrometry (MS) system was adopted to scrutinize the fluctuations in LiH content within the silicon solid-electrolyte interphase (SEI) during lithiation and de-lithiation processes. Here, the lithiation and de-lithiation degree of silicon in the half-cell can be precisely controlled by setting different cutoff voltages. Titrated test samples (silicon electrodes) were obtained by disassembling.

silicon /Li half-cells at various lithiation and de-lithiation degrees. We verified the reactions of deuterium oxide ($D_2O$) with lithium-silicon alloy and LiH, resulting in the generation of $D_2$ and HD gases, respectively (Figure 2a). As depicted in Figure 2c and 2d, an incremental increase in the content of Li–Si alloy and LiH was observed during the lithiation process. The emergence of LiH during lithiation

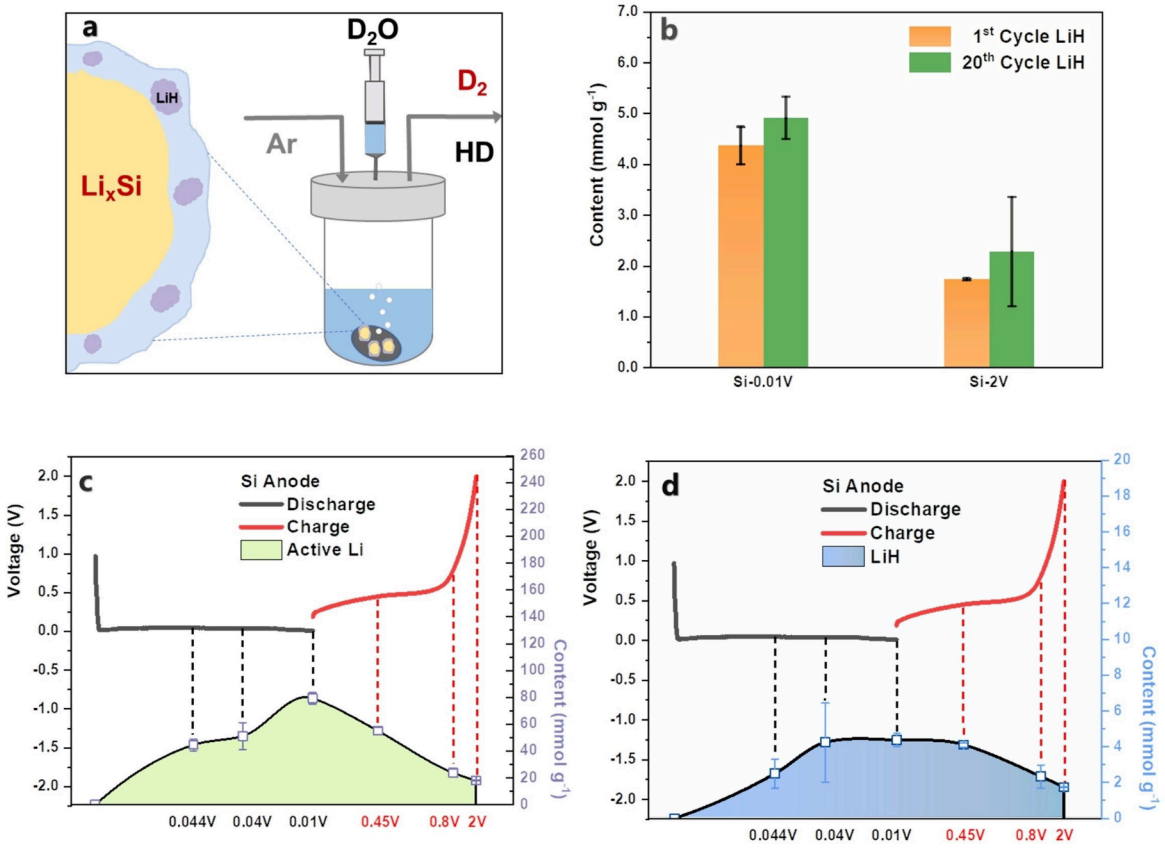



***Figure 2.*** **The evolution of LiH components in Si electrodes during cycling.** (a) Schematic illustration of online $D_2O$ titration gas analysis MS system. (b) The amounts of LiH in the SEI of cycled Si anode at the 1$^{st}$, and 20$^{th}$ cycle of a Li–Si half-cell, and the x-axis represents the voltage state when sampling. The evolution of Li(c) and LiH(d) upon the discharging-charging process of Li–Si half-cell. The drawing of the error bars in the figure is based on the electrode pieces in three independent battery after cycled ($n=3$).


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can be ascribed to side reactions transpiring at the electrode/electrolyte interface. Intriguingly, during the de-lithiation process, the variation trend of LiH deviates from that of the Li–Si alloy especially when the voltage rises from 0.01 V to 0.45 V (Figure 2d). In the preliminary stage of Li–Si alloy de-lithiation, the content of LiH decreases slowly. And a precipitous decline in LiH accumulation is observed during the later stage of de-lithiation. The dynamic evolution of LiH during the lithiation/de-lithiation process of the silicon electrode remains visible even after 20 cycles, signifying the ongoing reaction of LiH throughout cycling(Figure 2b). And the significant increase in the accumulation of LiH on the delithiated silicon anode after 20 cycles can be atributed to the repeated regeneration and accumulation of SEI due to the volume changes associated with silicon cycling.

Drawing from prior studies, the de-lithiation of silicon can be primarily divided into two stages. Initially, as the potential increases, the silicon in the electrode transitions from isolated Si anions (a-$Li_xSi$, $x=3.5$–$3.75$) to small Si clusters (a-$Li_xSi$, $x=2.0$–$3.5$). During this stage, the content of LiH shows negligible change. However, upon reaching an electrode potential of 500 mV, a stark decrease in LiH content is noted. At this point, the electrode composition consists of large Si clusters and Si extended networks (a-$Li_xSi$, $x=0$–$2.0$). We hypothesize that the abrupt decrease in lithiated hydrogen content at 500 mV is likely correlated with the pronounced emergence of large Si clusters, and thus propose the following reaction pathway:

$$xLiH + Si \rightarrow Li_xSi + x/2H_2$$

To validate this reaction, LiH and Si were mixed through a ball milling process (Figure 3a). The changes in LiH and Si before and after the reaction were monitored using mass spectroscopy titration. As shown in Figure 3a, the active lithium content (corresponding to the $D_2$ signal) in the ball-milled LiH and micron silicon mixture significantly increased, while the pure LiH showed no significant change in active lithium content after ball milling. This confirms the occurrence of the lithiation reaction of LiH with silicon. Furthermore, we separately assembled full solid-state batteries with NCM811-Si and NCM811-Si-LiH (mixed with LiH) and observed that the experimental group of batteries (Figure 3b), which introduced LiH into the anode throughout the battery cycling process, consistently exhibited

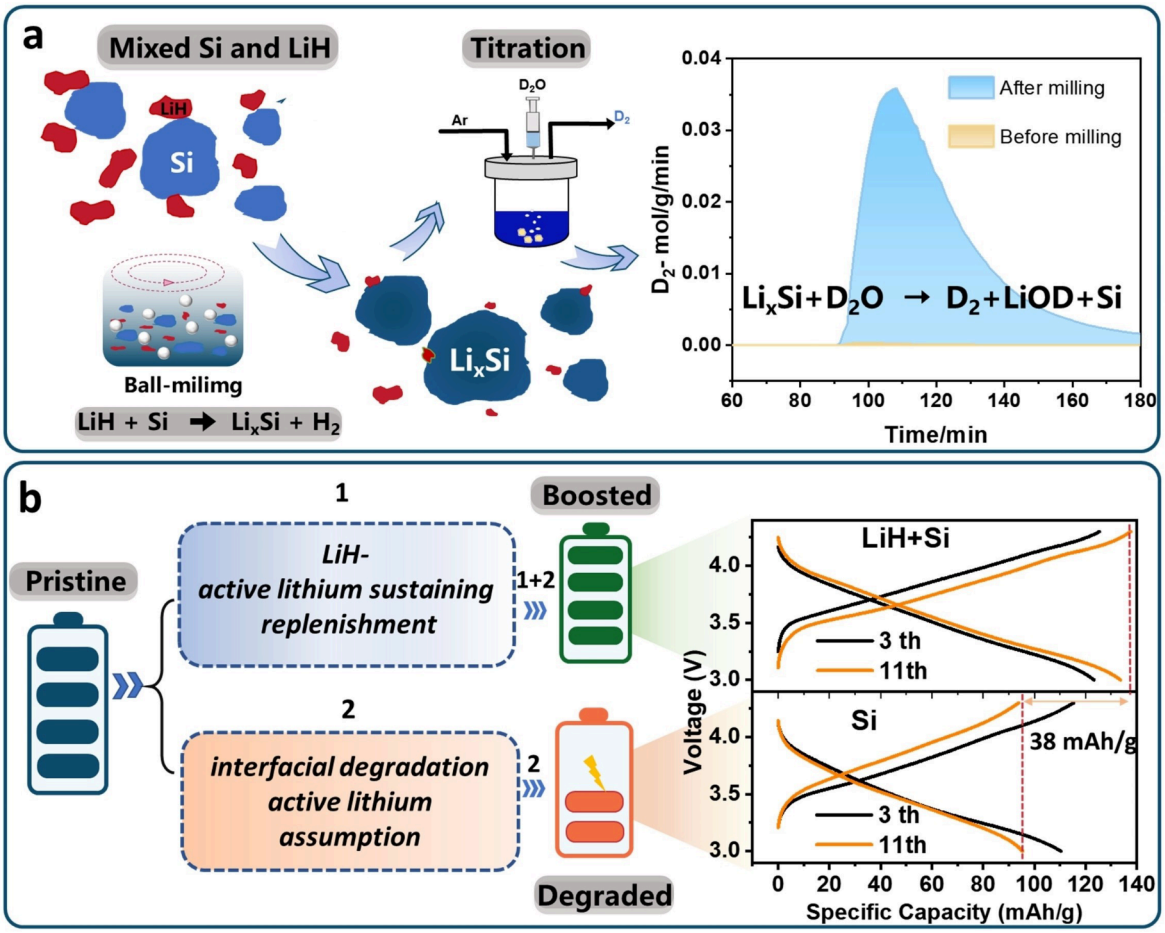



***Figure 3.*** **The verification of the reaction between Si and LiH.** (a) The $D_2$ evolution rate curve after $D_2O$ titration on samples of LiH and the ball-milling sample (the mix of LiH and Si). (b) The Voltage-Specific capacity curve of the NCM 811-LiSiPSCl-Si and the NCM 811-LiSiPSCl-Si (mixed LiH) in different cycles, and the schematic diagram of continuous replenishment of active lithium by LiH in all-solid-state battery.


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significantly higher capacity than the pure silicon, both at the 3rd cycle (0.3 C) and 11th cycle (0.1 C).

After 11 cycles, the specific capacity of the pure silicon negative electrode decreased by 13 mAh/g. In contrast, the experimental group with LiH addition maintained a higher specific capacity consistently, which even exceeded that of the pure silicon group at the 11th cycle, reaching 38 mAh/g. It is worth noting that one of the key reasons for selecting a solid-state system to validate this experiment is the notable and substantial side reactions that occur between the liquid electrolyte and LiH due to its highly reducible nature. Attempts conducted using liquid systems have failed to achieve successful charging and discharging due to these side reactions. Under consistent N/P ratios and other experimental conditions, the continuously higher capacity throughout the battery cycle indicates the ongoing lithiation reaction of LiH with silicon in the all-solid-state system (Figure S2), realizing the sustainably replenish of active lithium, which is in good agreement with the titration results. Despite belonging to the IV A, compared with Si, significant differences have been revealed in the reaction mechanisms between LiH and Ge alloy anodes, where previous studies have found that LiH reacts with Ge to form $Li_4Ge_2H$, and the accumulation of insulating $Li_4Ge_2H$ upon cycling seriously retards the charge transport process and ultimately triggers the Ge anode failure.

Upon dismantling the solid-state batteries after a single cycle, we found that the silicon electrode in the LiH group exhibited noticeable pulverization (Figure S3), suggesting that LiH in the silicon electrode's SEI likely plays a role in exacerbating its pulverization. We employed high-resolution transmission electron microscopy (HRTEM) to observe the phase information of lithiated silicon obtained through electrochemical charging, as well as lithiated silicon chemically lithiated with LiH. By examining diffraction spots from different regions, it was observed that electrochemically lithiated silicon exhibited uniform lithiation degree (Figure S4), whereas LiH lithiated silicon showed extreme non-uniformity with adjacent regions on the same particle displaying pure silicon phases, $Li_{12}Si_7$, and $Li_{22}Si_5$(Figure S5). One possible reason is that the local point contact between LiH and silicon compresses the unreacted region of silicon particles, significantly restricting the ionic diffusion at the reaction front under compressive strain, thereby preventing further reaction and resulting in the observed uneven degree of local lithiation in the experiment.[11] It is well known that internal stress and strain in silicon-based materials are closely related to the depth of lithiation. Greater lithiation depth leads to increased stress and strain. Based on the observed extremely non-uniform local lithiation phenomenon, we reasonably infer that the highly non-uniform stress and strain induced by LiH accelerates particle fracture and pulverization of silicon. Accordingly, we consider this as an important factor contributing to the pronounced pulverization of the silicon electrode in the LiH group, as observed after dismantling the solid-state battery following a single cycle (Figure S3).

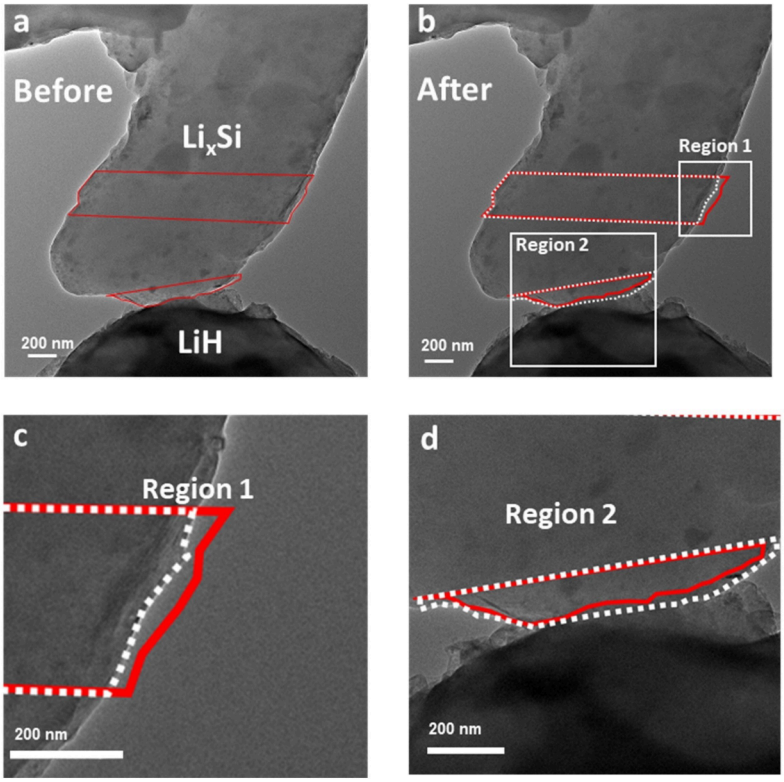



***Figure 4.*** **The verification of the reverse lithiation of LiH.** TEM images of the $Li_xSi$-LiH–Li nano battery before in situ charge(a) and after in situ charge(b). (c) TEM image of the $Li_xSi$ from a white boxed region (1) in (b). (d) TEM image of the $Li_xSi$ from a white boxed region (2) in (b).


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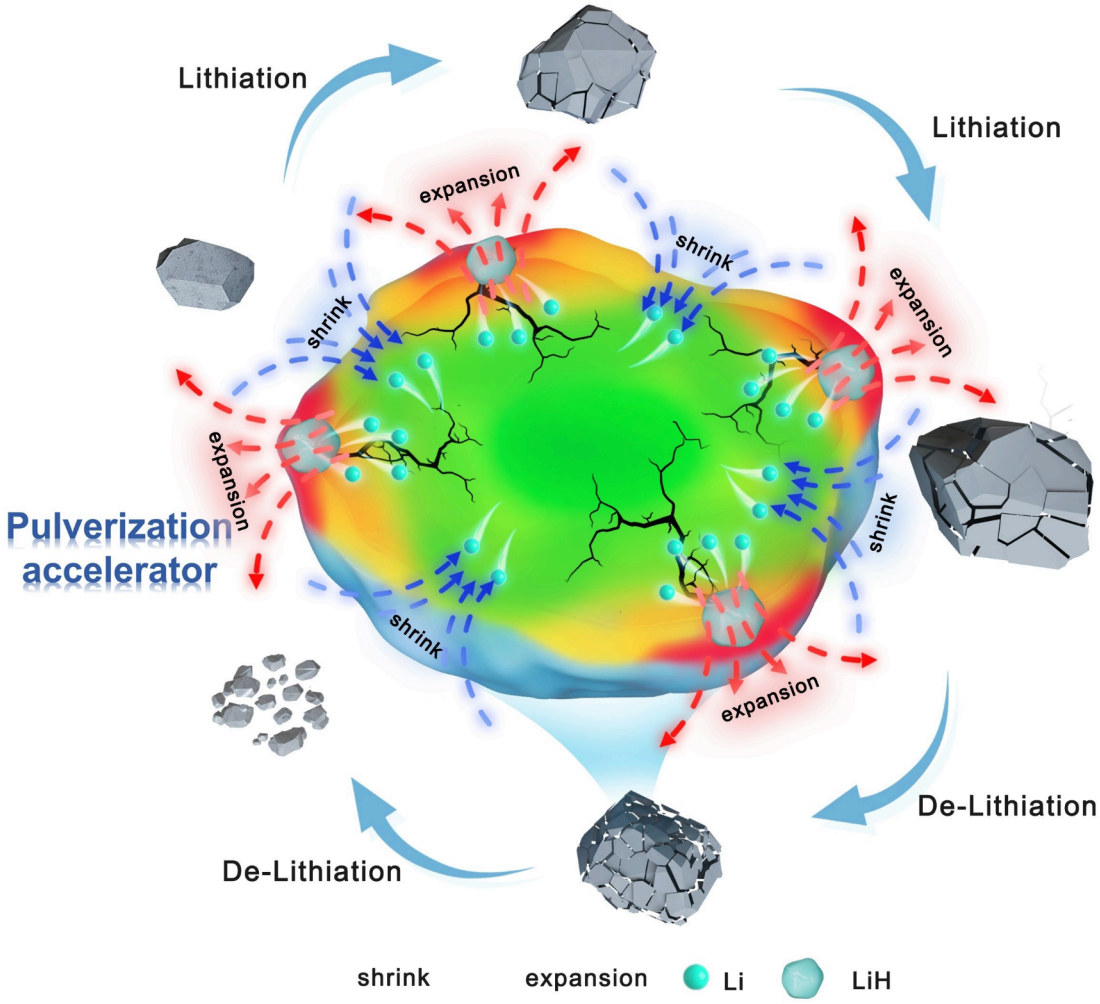



***Figure 5.* Schematic diagram of the LiH promoting particle pulverization**. The reverse lithiation behavior of LiH on micron silicon during de-lithiation exacerbates the localized stress in silicon particles and contributes to particle pulverization.

To further investigate our hypothesis, we designed a nano-battery, $Li_xSi$–LiH–Li, utilizing the in situ transmission electron microscopy (TEM) technique (as shown in Figure 4a&S6). The aim was to observe the influence of LiH on the morphological evolution and stress distribution of micro silicon during the lithium extraction process. The study revealed that the lithium-silicon alloy experienced volume contraction in regions away from the LiH (Figure 4, red solid line box represents the state before de-lithiation, while the white dashed line box represents the state after de-lithiation,). However, in the micro-regions in contact with LiH, silicon exhibited counterintuitive volume expansion.

It is widely known that the degree of lithiation is directly proportional to the volume change in silicon. The lithium extraction always results in volume contraction, of the lithium–silicon alloy. Therefore, the expansion observed in silicon near the LiH region provides evidence of its reversal lithiation after particle de-lithiation. The effect of local expansion results in an uneven stress distribution within the silicon particles. In fact, the majority of silicon particles exhibit intricate microstructures on their surfaces, characterized by irregularities, grooves, and microcracks. This complicated surface morphology, when subjected to localized expansion and stress induced by LiH, can easily exceed the critical stress level (stress concentration at the crack tip) of the material, thereby promoting crack propagation. Consequently, the material undergoes fracture and fragmentation (as shown in the finite element simulation in Figure S7). Based on these observations, we reasonably propose that LiH plays a role in facilitating silicon particle fragmentation (Figure 5), as the previous correlation we observed between the content of LiH in the anode and the degree of pulverization in silicon-based batteries. It is well known that pulverization easily triggers the loss of electrical contact in active electrode materials, and with repeated cycles, the process of battery degradation will be noticeably accelerated. Additionally, as a highly abundant inorganic component within the SEI, the instability of LiH can directly lead to structural instability of the SEI. When LiH is consumed, the resulting gap will compromise the isolation and protective functions of the SEI, leading to localized SEI breakdown. Consequently, the interface instability caused


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by LiH can result in additional loss of active lithium and electrolyte. Therefore, we reasonably suggest that researchers can optimize the cycling stability of silicon-based batteries by regulating the LiH content in the SEI through electrolyte design.

## Conclusion

As a critical component of SEI, the role of LiH in the morphological evolution of micron silicon anodes was firstly investigated using multidimensional and multiscale characterization techniques. A novel mechanism in which LiH-induced localized lithiation of micron silicon leads to uneven stress distribution and potentially accelerates micron silicon pulverization was firstly revealed qualitatively, providing new evidence for a strong association between intrinsic SEI instability and micron silicon anode failure, and offering new design principles for future development and advancement of SEI on high energy density and high stability silicon anodes.

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## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

**Keywords:** Si anode failure mechanism · solid electrolyte interphase · lithium hydride evolution · lithium-ion battery · pulverization

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